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Single Qubit Quantum Systems

Multiple Qubit Quantum Systems: Direct/Tensor Product, Entanglements, Gates and Quantum Parallelism

Lab: Interactive Notes on Multi-Qubit Quantum Systems

1h

Video: Multi Qubit Quantum States

29 min

Graded Assignment: Multiple Qubit Quantum States

Submitted

Video: Multi Qubit Quantum Gates

28 min

Graded Assignment: Multiple Qubit Quantum Gates

40 min

Video: Quantum Parallelism

14 min

Video: No Cloning Theorem

5 min

Graded Assignment: Quantum Parallelism and Implementing Classical Gates

45 min

Bell's Inequality and An Entanglement Game

Grover's Search Algorithm

Problem Set: Week 2

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%

Multiple Qubit Quantum Gates

Next item →

1. Suppose we have a quantum system in the state $|00\rangle$ and we wish to transform it into the state $\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$ 3 / 3 points

Review Learning Objectives

Assignment details

- Select all the valid unitary operations to achieve this.
- ☐

Make two copies of the state and apply Pauli-X gates to the second qubit on both copies but only to one of the copies of the first qubit and then "mix them up" in a uniform superposition.
- ☐

Apply a Pauli-X gate to the first qubit and a Hadamard gate to the second qubit.
- ☐

There is no unitary transformation that can achieve this.
- ☒

Apply a Hadamard operation to the first qubit and a Pauli-X (quantum not) gate to the second qubit.

100%

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See feedback

2. Suppose we have a three qubit system in the uniform superposition: $|\psi\rangle = \frac{1}{\sqrt{8}}(|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle)$ 3 / 3 points

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We wish to transform it to $|\psi'\rangle$ given by

$$|\psi'\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |010\rangle)$$

Which of the following operations will achieve this?

- ☒

Apply a Hadamard gate to the first and third qubits but leave the second qubit unaltered.
- ☒

Correct
Correct: Hadamard gate is its own inverse. Applying to a uniform superposition of a single qubit will yield the state $|0\rangle$ back.
- ☐

Zap the components of the superposition corresponding to $|001\rangle, |111\rangle, \dots$ that are not required in the desired result and we are done.
- ☐

There is no way to do this with a unitary operation.
- ☒

Apply the operator $H \otimes I \otimes H$ where H stands for the single qubit Hadamard gate while I is the single qubit identity operator.
- ☒

Correct
Correct: this is another way of stating "Apply a Hadamard gate to the first and third qubits but leave the second qubit unaltered."

3. How do we transform a state $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ 2 / 2 points

into the state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$?

- ☒

Apply a controlled-X (quantum not) operation to the second qubit with the first qubit as the control qubit.
- ☐

Measure the first qubit and if we obtain a $|1\rangle$ apply a X gate on the second qubit. If we obtain a $|0\rangle$ leave the second qubit alone.
- ☐

Apply a X gate (quantum not) operation to the second qubit.
- ☐

Apply a controlled-X (quantum not) operation to the first qubit with the second qubit as the control.
- ☒

Correct
Correct: that will do it.

4. Suppose we have the superposition $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ 2 / 2 points

What is the result if we apply a controlled phase gate with phase $\frac{\pi}{2}$ on the second qubit with the first qubit as the control?

- ☒

 $\frac{1}{\sqrt{2}}(|00\rangle + i|11\rangle)$
- ☐

 $\frac{1}{\sqrt{2}}(i|00\rangle - |11\rangle)$
- ☐

 $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

Correct
Phase of $\pi/2$ multiplies the second component by $e^{i\pi/2} = i$.